Lesson 13B Gravitation Near Earth's Surface

Gravitation Near Earth's Surface

To determine the **gravitational acceleration** near Earth's surface, we equate Newton's law of gravitation with Newton's second law:

$$F = \frac{GM_E m}{r^2} = ma_g$$

Solving for the gravitational acceleration a_g:

$$a_g = \frac{GM_E}{r^2}$$

Where:

- M_E is the mass of the Earth
- r is the distance from the center of the Earth
- G is the gravitational constant

This gives the **magnitude of gravitational acceleration** at any distance from Earth's center. Refer to the following table for values of a_g at various altitudes.

Table 13.2.1 Variation of a_g with Altitude

Altitude (km)	a_g (m/s^2)	Example
0	9.83	Mean Earth surface
8.8	9.80	Mt. Everest peak
41.4	9.70	Highest parachute jump
400	8.70	International Space Station
35 700	0.225	Communication satellite

Measured vs. Calculated Gravity

The calculated value of a_g may differ slightly from the locally measured gravitational field strength g due to:

1. Non-uniform mass distribution within Earth (see associated figure)
2. Earth's shape as an oblate spheroid, not a perfect sphere
3. Rotational motion, particularly significant near the equator

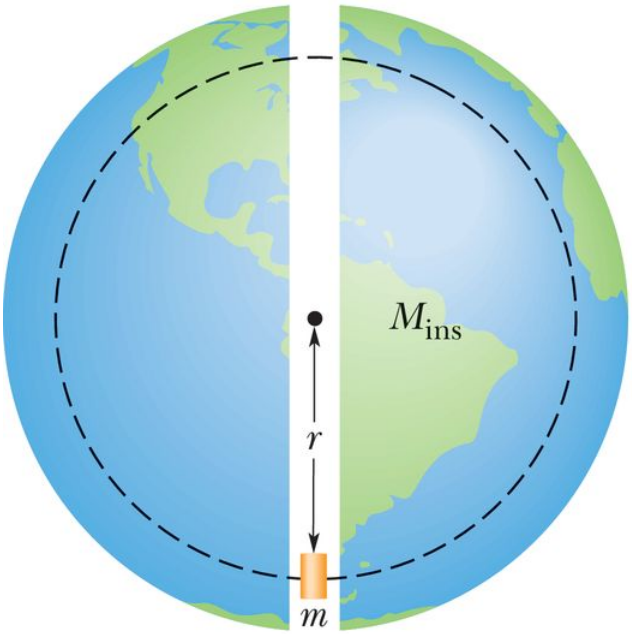
Example: Apparent Weight at the Equator

At the equator, an object's **apparent weight** is slightly reduced compared to the gravitational force due to **centripetal acceleration** from Earth's rotation. This difference highlights why weight and gravitational force are not always identical.

Gravitation Inside Earth

According to the **shell theorem**:

- A spherical shell exerts no net gravitational force on a particle inside it
- For a solid, uniform-density sphere, the gravitational force inside increases linearly with distance from the center



Assuming constant density, the Earth's average density is:

$$\rho = \frac{M_E}{\frac{4}{3}\pi R^3}$$

The mass enclosed within a sphere of radius r (denoted M_{ins}) is:

$$M_{\text{ins}} = \rho \cdot \frac{4}{3}\pi r^3$$

Substituting into Newton's law:

$$F = \frac{GM_{\text{ins}}m}{r^2} = G\frac{m}{r^2} \cdot \rho \cdot \frac{4}{3}\pi r^3 = \left(\frac{4}{3}\pi G\rho\right) mr$$

$$\Rightarrow F = Kmr \quad \text{where} \quad K = \frac{4}{3}\pi G\rho$$

This equation has the same form as **Hooke's law** ($F = -kx$), which means an object falling through a tunnel bored through Earth would **oscillate harmonically** about the center of the planet.